

Testing Significance of Moderator Effects on Multiple Subject Behavior Networks

In psychological research, there is a pressing interest in understanding how socioeconomic status, physical health status, and prior mental health diagnoses may influence the emotional and behavioral patterns of individuals. A deeper understanding of these influences can aid in the development of comprehensive psychological profiles for subject assessment, and can ultimately inform personalized psychological interventions [1,2]. Motivated by these analytical needs, the goal of this project is to develop and make available a toolkit of novel statistical models and efficient algorithms to quantify and test the impact of between-person differences in the behavioral patterns of individuals over time.

This project is motivated by the investigators' collaborative project between the Departments of Psychology and Statistics at UCR [3], where we aim to use multi-subject longitudinal data to estimate Granger-causal networks that describe how depression, anxiety, stress, quality of sleep, and physical activity influence each other. An example of multi-subject Granger-causal networks is found in Figure 1. Nodes in the networks correspond to behavioral constructs, and links describe Granger-causal effects, *i.e.*, whether past values of one construct predict future values of another [4]. For example, a directed link from sleep quality to anxiety suggests that poor sleep quality will have a negative effect on next day's anxiety levels. Each of these links can potentially be affected by *moderators*, which are additional constructs that influence the strength of relations of the Granger-causal effects.

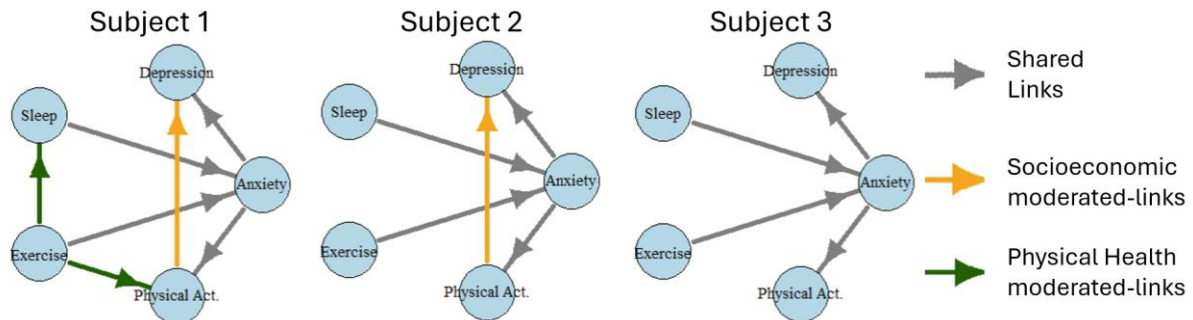


Figure 1: Example of Granger-causal networks across 3 subjects. Each subject has a different behavioral network. Our aim is to quantify whether different network links are moderated by external socioeconomic status or health status.

Prior methods in the literature can perform the estimation of such Granger-causal networks on multi-subject longitudinal data [5,6,7,8,9], and test network significance for a single subject [4,10,11,12]. However, these methods either do not incorporate moderator information [4,5,6,7,8,9,10,11,12] leading to false discoveries, or cannot test whether the estimated networks and moderator effects are statistically significant [5,6,7,8,9], leading to potentially biased results. The development of these tests is challenging because longitudinal observations are highly correlated, reducing the effective sample size [4]. Additionally, analyzing heterogeneous subjects increases model complexity, with the number of parameters growing combinatorially with the number of variables. For example, for 10 moderators, 10 variables in the Granger-causal network, and 50 subjects, we must assess a total of 22,500 parameters. Performing reliable inference in these high-dimensional settings with correlated observations is a substantial statistical and computational challenge.

In our collaborative project at UCR, we have developed the moderator vector autoregressive model (MOD-VAR), a new method that estimates individual Granger-causal networks and quantifies the effect of moderator variables on the network links of each subject [3]. MOD-VAR is effective in high-dimensional settings as confirmed via theoretical guarantees and extensive numerical experiments. This work is in preparation for submission to *Psychological Methods*.

Objectives

The core objective of this proposal is to develop a statistically rigorous hypothesis testing framework based on the MOD-VAR to assess, for each potential network edge and moderator variable, whether the moderator significantly influences Granger-causal effects across subjects. The effectiveness of our testing procedure will be verified through theoretical analysis, numerical experiments, and real-data applications. We aim to develop software implementation in R, making this toolkit widely accessible for analyzing multi-subject longitudinal data.

In the future, we aim to expand MOD-VAR beyond behavioral longitudinal studies through collaborations with interdisciplinary research teams at UCR and other institutions. We also plan to extend the theoretical idea to other statistical frameworks. Furthermore, we plan to apply for extramural funding to support the research activities of these extensions.

Specific aims

[A1] Develop the significance-testing procedure [Months 1-6]: Develop a multiple testing procedure controlling the false discovery rate across moderator-network edges, accounting for high dimensionality and subject heterogeneity.

[A2] Implement the method in R [Months 7-12]: Implement the testing method and release it as an R package.

[A3] Evaluate performance [Months 7-12]: Assess the testing procedure on simulated and real data. Apply the procedure to the GLOBEM dataset [13] as well as investigator-owned datasets to examine how socio-economic moderators influence behavioral patterns over time.

[A4] Explore extensions and external funding [Months 1-12]: Establish new interdisciplinary collaborations and prepare proposals for extramural funding.

Anticipated Results

We anticipate the development of the manuscript “*Testing Moderator Effects on Multiple-Subject Granger-Causal Networks on Intensive Longitudinal Data*,” together with a companion R package. In addition, we expect the establishment of new collaborative teams to extend MOD-VAR to general multi-subject longitudinal studies. One direction is to extend MOD-VAR for the analysis of brain connectivity networks using fMRI data to determine how moderators influence brain connectivity pathways [14,15,16]. A second direction is the analysis of multi-subject electroencephalogram (EEG) recordings to determine how brain activity varies across individuals [17,18,19,20,21]. Extramural funding will likely be required for the success of these innovative extensions. The UCR Research Small Grant will provide early support for the development of our MOD-VAR methodology and generate preliminary results vital for future grant applications and the long-term advancement of our research program.

References

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Budget Justification

Personnel - \$10,887 USD.

To Be Named (TBN), Graduate Student (50% effort for 3 summer months).

Salary support for the GSR to assist in the development, implementation and evaluation of the MOD-VAR hypothesis testing procedure.

- **GSR 1st month of 50% summer effort (\$3,629 USD):** Develop, in coordination with the P.I. and co-PI, the method for testing moderator effects on behavioral networks, in accordance with Aim [A1].
- **GSR 2nd month of 50% summer effort (\$3,629 USD):** Implement the MOD-VAR testing procedure in R (Aim [A2]).
- **GSR 3rd month of 50% summer effort (\$3,629 USD):** Design simulations and conduct real data analyses on the HPCC cluster, based on previous simulation work (Aim [A3]).

Computational Resources – \$3,500 USD.

- **High Performance Computing Center (HPCC) subscription (\$2,000 USD):** Support for the subscriptions of the PI and co-PI's laboratories to the UCR HPCC for 12 months (June 2026 – July 2027), essential for Aims [A2, A3].
- **GSR Computer Equipment (\$1,500 USD):** Purchase of computer equipment to enable GSR to work on Aims [A2, A3].

Coordination and Travel - \$2,000 USD.

Funds to be allocated for PI and co-PI to visit funding agencies, and for university visits to coordinate new collaborations (Aim [A4]).

Total Direct Costs – \$16,387 USD.

Budget Summary

Cost Element	Explanation	Total
1 Personnel		\$10,887 USD
GSR Summer Support	Funding for 3 summer months to support GSR research activities.	\$10,887 USD
Computational		
2 Resources		\$3,500 USD
Computing Center Subscription	Funding to cover the annual subscription of the PI and co-PI's laboratory to the High-Performance Computing Center (HPCC) at UCR for the academic year of 2026-2027.	\$2,000 USD
Computer Resources for GSR	Funding to purchase computer equipment for the hired GSR.	\$1,500 USD
3 Coordination and Travel		\$2,000 USD
Coordination and Travel	Cover the fee and travel costs to visit funding agencies in order to increase success rate of application to extramural funding and coordinate collaborations with interdisciplinary teams.	\$2,000 USD
Total	(Direct costs)	\$16,246 USD